Irfan Tayyab

Full-Stack AI Engineer

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Education

BS in Computer Science, Virtual University of Pakistan

Final Year Project: an AI platform (FastAPI, LangChain, n8n) that routes students' queries from the web, email and WhatsApp **in a median of 1.2 seconds**, with model fallback, personal data masked from the models, and low-confidence queries passed to a human. Built with a coding-agent harness.

Work Experience

Software Engineer, AmentoTech Pvt Ltd — Nov 2025 – Oct 2026

- Architected a **configuration-driven form system** powering 8+ multi-step variations, with schema branching and per-step validation, making new flows quick to add and easy for the team to build on together.
- Set up a Turborepo monorepo with shared linting, formatting and pre-commit checks (lint-staged), giving the team and its AI tools one consistent codebase.
- Created shared, reusable components and complex layouts with React Portals, turning Figma designs into pixel-perfect UIs with AI-assisted tooling and **minimal designer-developer revision cycles**.
- Engineered a widget dashboard with **per-user layouts** on a drag-and-drop grid (dnd-kit), letting users rearrange, resize and reshape widgets while the layout holds on every screen size.
- Adopted AI-assisted development with Claude and Cursor, using prompt engineering to streamline the work, alongside AI-powered pull request reviews and the team's custom skills plugin.

Open Source Contributor, Plumb MCP — Aug 2026

- Fixed a Figma **UI freeze** in Plumb MCP's plugin bridge by moving selection sync from push to on-demand pull, removing a full node-tree walk on every click. Reused the bridge's existing request/reply path. Merged upstream with all 955 tests passing (commit ↗).

MERN Stack Engineer, Sprouto Group — Mar 2025 – Nov 2025

- Developed a **RAG-pattern AI chatbot** over uploaded documents (LlamaParse, Supabase) that helped teachers catch plagiarism in student submissions. Tested with real users.
- Shipped Brandably's B2B merchandise store end to end (Next.js, Stripe, Postgres), with pipelines that import a supplier's full catalog and sync its prices and stock automatically. **Now live and taking orders.**
- Led full-stack development within strict deadlines, delivering the corporate website with SEO-friendly server-side rendering and scroll-based parallax animations.
- Designed a typed API integration layer with centralized error handling, plus reusable React components.
- Managed the database, file storage and client-side server-state caching, cutting load times on slow connections.
- Migrated 1.9 GB of data from MySQL on AWS RDS to PostgreSQL on Supabase, and 3 GB of files from AWS S3 to Supabase Storage, with **zero data loss**.

MERN Stack Developer, CodeFlamme Pvt Ltd — Jul 2024 – Feb 2025

- Built a real-time chat system and live notifications with Socket.io (WebSockets).
- Implemented one-time payments and auto-renewing subscriptions with Stripe and PayPal, plus a referral system.
- Secured routes with role- and region-based authorization, plus Google, LinkedIn and Facebook OAuth sign-in.
- Localized the frontend with i18next and established a multi-language transactional email system on the backend.

Skills

Languages	Python, TypeScript, JavaScript, SQL, HTML, CSS
Frontend	React.js, Next.js, Redux Toolkit, TanStack Query, React Hook Form, Zod, Tailwind CSS, Material UI, Figma
Backend	FastAPI, SQLAlchemy 2 (async), asyncpg, Pydantic v2, Alembic, uv, Ruff, Pyright (strict), Node.js, NestJS, GraphQL (Apollo), REST APIs, WebSockets (Socket.io), JWT, n8n, Gmail API, WhatsApp
Databases	PostgreSQL, MySQL, MongoDB, Prisma, Drizzle, Supabase, Firebase
AI Engineering	LLM APIs (Gemini, Groq, Claude), LangChain, Model Context Protocol (MCP), Structured Output, Model Fallback, RAG, PII Redaction, NLP (Intent Detection, Entity Extraction, Text Classification)
DevOps & Testing	Docker, Caddy, AWS, GCP, Oracle Cloud, Vercel, GitHub Actions, Git, Jest/Vitest, React Testing Library, pytest
AI Workflow	Prompt Engineering, Context Engineering, Coding-Agent Harness Engineering, Agentic Coding, Spec-Driven Development